Siyang Liu

7348347522 • lsiyang@umich.edu • Ann Arbor, USA • Google Scholar • Linkedin • Personal Page

SUMMARY

PhD candidate in Computer Engineering at the University of Michigan with expertise in large language models, human participant simulation, and web automation. Skilled in building end-to-end systems spanning tokenization, post-training, generative agents, RAG workflows, and web agents. Seeking to advance AI that bridge human-centered design with scalable, real-world impact.

EDUCATION

University of Michigan — **Ann Arbor, MI**
Doctor of Philosophy in Computer Engineering — *May, 2027*
GPA: 4.00/4.00

Tsinghua University — **Beijing**
Master of Computer Science and Technologies — *June, 2021*

SKILLS

*Large Language Models (LLM):*In-context learning, instruction-tuning, preference optimization, RAG, human-in-the-loop post-training.
User Simulation & Agents: Role/agent design, user /participant modeling, multi-agent systems, character synthesis
UI Automation: Web agent framework development (independent), UI automation, full-stack system design
Programming & Systems: Python, Java, JavaScript, C++, concurrency, full-stack development, data processing

RESEARCH EXPERIENCE

University of Michigan/ Language and Information Technology Group — **Ann Arbor, MI**
Graduate Student Research Assistant — *2022-Present*
Advisor: Rada Mihalcea

- Designed a **patient-centered RAG system** for oncology visit preparation, improving usability (UMUX = 6.0/7), content relevance (6.7/7), and clinical faithfulness (6.82/7); submitted to EMNLP 2025 Demonstration, demo link.
- Developed an **expert-in-the-loop** depression simulation model, combining **supervised fine-tuning and preference optimization**; produced a realistic 8B LLM outperforming OpenAI baselines in linguistic authenticity and profile adherence (ACL Findings 2025), project link.
- Led an **age bias study** across six LLMs to characterized their bias toward youth values; published at EMNLP 2024.
- Engineered **task-adaptive tokenizers** to enhance long-form generation efficiency by up to 60% while maintaining output quality; published at EMNLP 2023.

Tsinghua University/Conversational AI Lab — **Beijing**
Graduate Student Research Assistant — *2019-2021*
Advisor: Minlie Huang

- Rectified the widely used generation diversity metric Distinct by correcting its mathematical bias toward longer outputs, achieving stronger correlation with human judgment; published at ACL 2022.
- Built emotional support dialogue systems and benchmarked the emotional support conversation task, pioneering work in AI for mental health; published at ACL 2021.

WORK EXPERIENCE

Educational Testing Service — **Princeton, NJ**
Research Intern — *June, 2025*

- Developed a student-role driven **web agent** system to simulate digital assessment experiences, accelerating testing and prototyping cycles for educational platforms.
- Delivered internship presentation showcasing early progress toward scalable, AI-powered system evaluation.

Tencent America — **Bellevue, WA**
Research Intern — *May, 2024*

- **Developed crowdsourcing framework** treating LLM agents as recruitable, screenable, and engageable participants across UX research stages; Released open-source toolkit with modular pipeline and **curated billion-scale profile pools** to enable adoption and reuse; **Validated at scale** in a game prototyping study: 240 simulated players from 2,900 profiles; aggregated agents reproduced ~90% of human findings. One local human ≈ 12.8 agents; one crowdsourced human ≈ 3.2 agents. Submitting to CHI 2026.

Kwai (快手) **Beijing, China**

Algorithm Engineer *July, 2021*

- E-commerce merchant recommendation based on neural graph network
- Achieved ~5% GMV increase in A/B testing and deployed as a recall strategy on E-commerce merchant recommendation business.

PROJECT EXPERIENCE

WebOasis (Independent Development) **Ann Arbor, MI**

Independent developer *August, 2025*

- Built **WebOasis (**github repo), an open agent framework for navigating complex web worlds — "Any site. Any page. Any UI. Any complexity."
- Added one-parameter engine switch (Playwright ↔ Selenium) and a dual-agent architecture for modularity and scalability.

PUBLICATIONS

1. **Liu SY**, An L, Mihalcea R, Patient-Centered RAG for Oncology Visit Aid Following the Ottawa Decision Guide, *EMNLP Demonstration Track (under review)*, 2025
2. **Liu SY**, Sabour S, Wang X, Mihalcea R, Free Lunch for User Experience: Crowdsourcing Agents for Scalable User Studies, *The Proceedings of the ACM on Human-Computer Interaction (under review)* , 2026
3. **Liu SY**, Brie B, Li W, Biester L, Lee A, Pennebaker J, Mihalcea R, Eeyore: Realistic Depression Simulation via Expert-in-the-loop Supervised and Preference Optimization, *ACL Findings*, 2025
4. Sabour S, Liu JM, **Liu SY**, Yao CZ, Cui SY, Zhang XM, Zhang W, Cao, YR, Bhat A, Guan J, Wu W, Mihalcea R, Althoff T, Lee T, Huang M, Human Decision-making is Susceptible to AI-driven Manipulation, *Nature Communications (under review)*, 2025
5. **Liu SY**, Maturi T, Shen SQ, Mihalcea R. The Generation Gap: Exploring Age Bias in the Value Systems of Large Language Models. *EMNLP, 2024*.
6. Sabour S, **Liu SY**, Zhang ZY, Liu JM, Zhou JF, Suharto A, Li JZ, Lee TMC, Mihalcea R, Huang ML. EmoBench: Evaluating the Emotional Intelligence of Large Language Models. *ACL, 2024*.
7. **Liu SY**, Deng NH, Sabour S, Huang ML, Mihalcea R. Task-Adaptive Tokenization: Enhancing Long-Form Text Generation Efficacy in Mental Health and Beyond. *EMNLP, 2023*.
8. **Liu SY***, Sabour S*, Zheng YH, Ke P, Zhu XY, Huang ML. Rethinking and Refining the Distinct Metric. *ACL, 2022*.
9. Sun H*, Lin ZR*, Zheng CJ, **Liu SY**, Huang ML. PsyQA: A Chinese Dataset for Generating Long Counseling Text for Mental Health Support. *ACL Findings, 2021*.
10. **Liu SY***, Zheng CJ*, Demasi O, Li Y, You Z, Huang ML. Towards Emotional Support Dialogue Systems. *ACL 2021*.
11. Ke P, Ji HZ, **Liu SY**, Huang ML. Sentilare: Linguistic Knowledge Enhanced Language Representation for Sentiment Analysis. *EMNLP, 2020*.

ACTIVITIES

Reviewer for Nature Communications and the Association for Computational Linguistics (40+ papers).

Organizer for NLP@Michigan Day and university-wide events including the AI Network and Building Bridges in CSE PhD Studies.